Sahaj Bhadja

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Summary

Software engineer with 2+ years building AI-enabled applications, secure APIs, automation workflows, and production data services. Experienced in Python, REST integrations, SQL, stateful orchestration, RAG, cloud platforms, CI/CD, testing, observability, and operational support.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built Python APIs, Azure SQL services, and operational dashboards for 500+ users, tuning indexed queries and secure access paths to reduce production response latency from 1.2 seconds to 350 milliseconds.
- Implemented stateful automation for 5,000+ monthly jobs using explicit transitions, triggers, retries, reconciliation, and failure isolation, tripling throughput while improving correctness, observability, maintainability, and production incident diagnosis.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed React interfaces and Flask REST services handling 10,000+ daily requests, implementing secure document ingestion, authentication, status tracking, and automated grading workflows while resolving production issues for 150+ users.
- Created concurrent processing for 1,000+ unstructured files with automated validation, AWS RDS persistence, structured logging, and failure handling, reducing turnaround 60% while maintaining reliable behavior under production load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built testable PyTorch simulation pipelines processing 100,000+ trajectory logs, improving long-horizon planning 18% through reproducible experiments, structured evaluation, model monitoring, and systematic debugging of intelligent system behavior.
- Developed a spatial-temporal encoder combining map geometry and agent context through multi-head attention, improving closed-loop stability 15% while documenting experiments and validating performance across diverse simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed Python services, SQL integrations, and event-driven ETL workflows using AWS Glue, S3, and REST APIs, improving response times 30% and data-processing efficiency 68% across secure production applications.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, accelerating incident diagnosis, code review feedback, and reliable production releases.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular React and Node.js features across eight services with JWT authentication and role-based access controls, synchronizing 2 million+ Azure SQL records through secure APIs and real-time WebSockets.
- Automated Docker builds, tests, staged deployments, and production monitoring through Jenkins and Azure, reducing release time 50% while improving code quality, deployment consistency, and operational visibility across environments.

Skills

Languages & AI: Python, Java, JavaScript, TypeScript, SQL, PyTorch, LLaMA-2, RAG, semantic search, vector databases, prompt engineering
Backend & Integration: FastAPI, Flask, Node.js, REST APIs, microservices, WebSockets, event-driven workflows, state management, triggers, orchestration, API integration
Data & Cloud: PostgreSQL, Azure SQL, SQL Server, MySQL, MongoDB, Firebase, AWS RDS, AWS, Azure, S3, Blob Storage, data modeling, query optimization
Quality & Security: Git, Docker, Kubernetes, Jenkins, CI/CD, PyTest, Mockito, automated testing, debugging, logging, monitoring, observability, authentication, RBAC, production support, Agile

Projects

Big-Brain | Live | *LLaMA-2, RAG, FAISS, PostgreSQL, REST APIs, Clerk*
- Built an authenticated RAG platform processing 50,000+ unstructured documents with LLaMA-2, FAISS, PostgreSQL, and REST APIs, combining semantic retrieval, conversational context, access controls, and real-time collaboration workflows.

HMIS | Live | *TypeScript, React, REST APIs, Appwrite, Sentry, Twilio*
- Built healthcare workflows for appointments, records, and consent-based access using React, REST APIs, authentication, notifications, and Sentry monitoring, supporting 100+ daily interactions while reducing administrative workload 30%.

FinNexus | Live | *TypeScript, React, REST APIs, WebSockets, MongoDB, Kubernetes*
- Built secure financial workflows supporting 50,000+ users and one million transactions, integrating REST APIs, real-time synchronization, granular filters, MongoDB persistence, monitoring, validation, and automated Kubernetes production deployment.